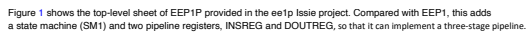
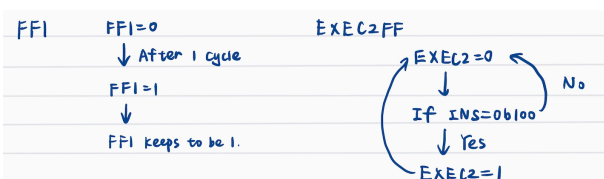


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```
testpipeline.txt
MOV RB, #7 1
LDR RA, [R0, #0] 2
LDR R1, [R0, #0] 2
SUB R2, #5, R8 // should be 8 3
ADD R0, #7 3
LDR R1, [R0, #0] 4
ADD R0, #7 4
MOV -1 5
| 6
7
```

Task 1. Load the statemachine sheet of ee1p to examine the design of SM1. The two flip-flops implement two independent state machines. Draw two separate state diagrams: one for FF1 and one for EXEC2FF.



Where b1000 is LDR, so only LDR starts EXEC2 and uses the stall design.

Using your FF state diagram, explain the INITV waveform and describe its effect on clock cycle 0 in Figure 3. EXEC2V is high during EXEC2: determine from your EXEC2FF state diagram when this occurs and check that this matches Figure 3.

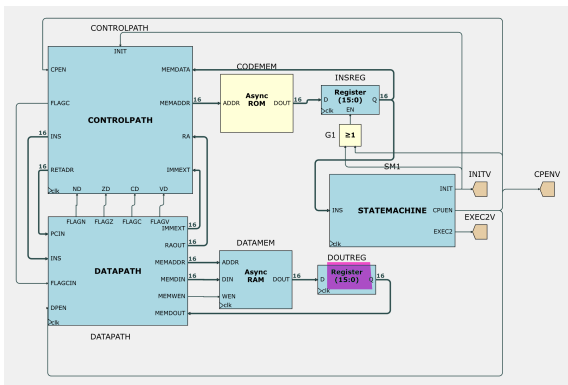
Machine cycle →	0	1	2	3	4	5	6	7
	N => not LDR or STR, L => LDR, S => STR							
FETCH	N1	N2	L3	N4	N4	S5	N6	
EXEC1		N1	N2	L3		N4	S5	N6
EXEC2				L3				
ADDR, WEN			L3			S5		
DIN						S5		
DOUT				L3				

Figure 3: EEP1P pipeline diagram



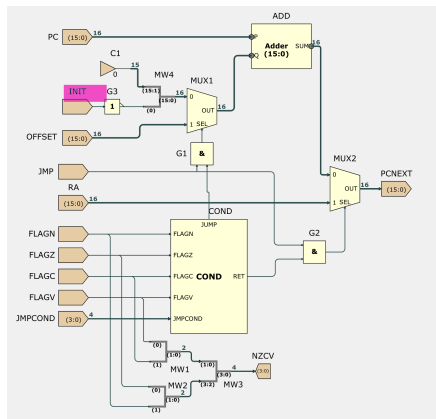
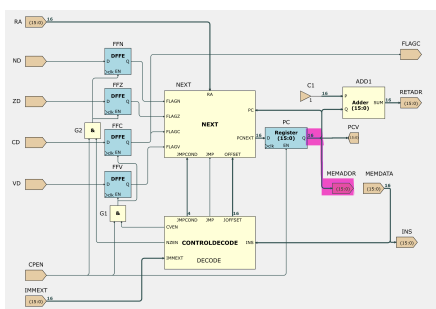
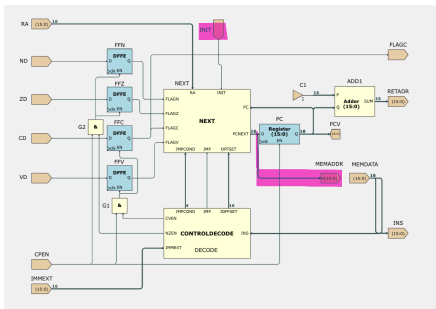
We can observe that EXEC2V is high in cycle 4, during which EXEC2 is on. That's because LDR requires 2 cycles to operate, so EXEC2 is operating and there is a stall in EXEC1 & FETCH.

INITV is only high in the first cycle as it only needs to tell CPU to start, as it is an initiate symbol.

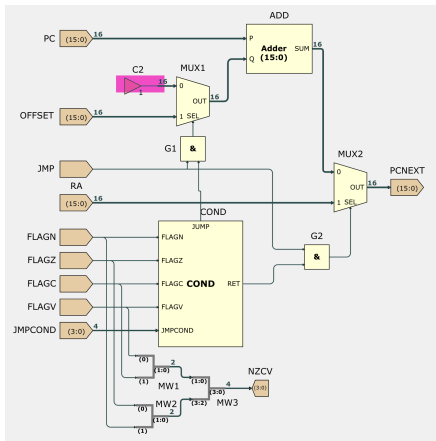


LDR instruction has to go through the register, so it needs 2 cycles to complete.

Task 2. Examine MEMADDR on the controlpath sheet of lab3p and compare it with the unpipelined lab2 EEP1 control path. Explain how the modification to MEMADDR allows the correct next instruction to be fetched.



Controlpath with pipeline



Controlpath without pipeline

We can see that, now if MUX1 selects 0, the input is inverse of INIT, which is 0 1 1 1... It gives an initial. While without pipeline, it's just 1, so now it leaves an initial state for pipeline to work.